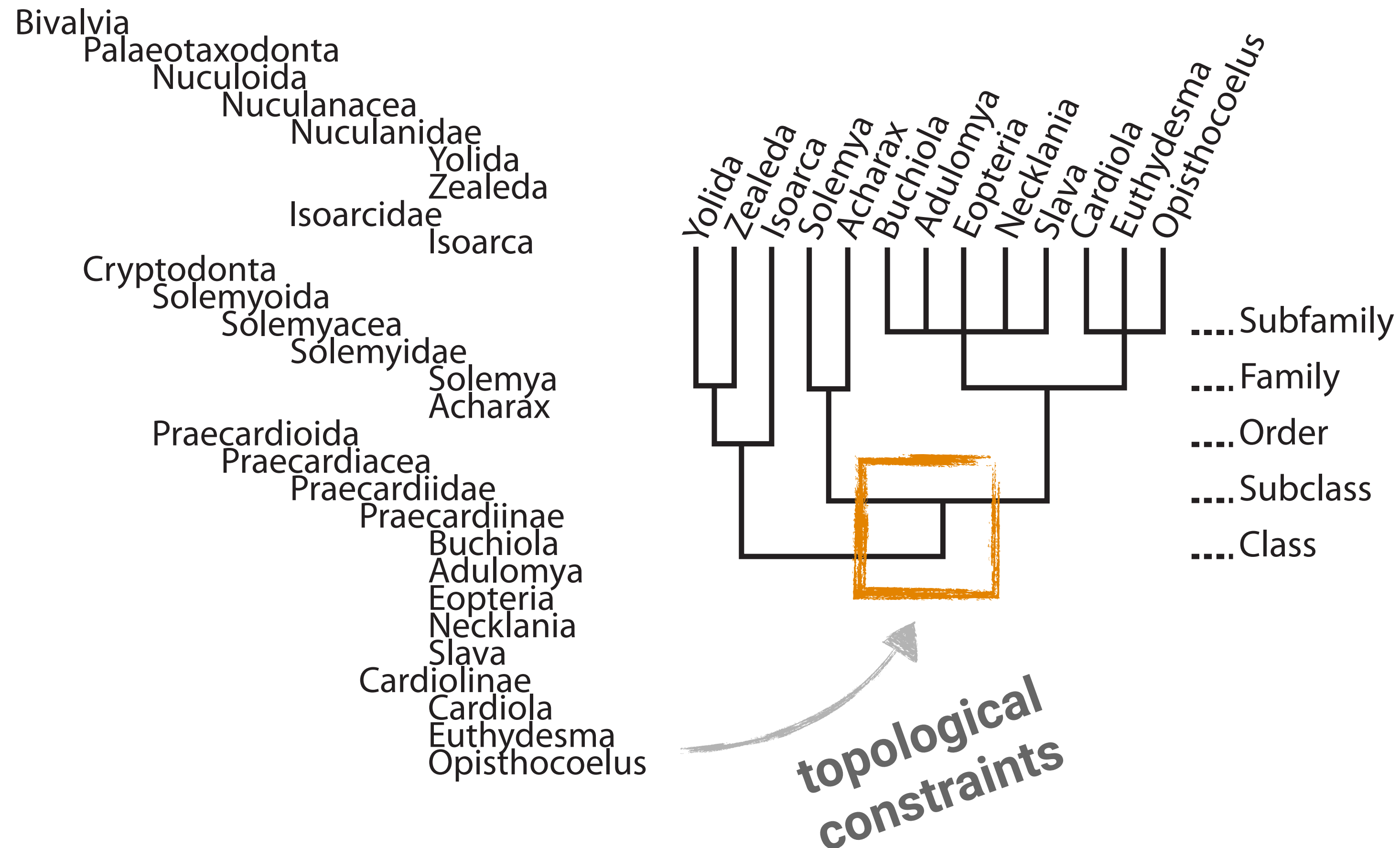


Exercise

Fossils can be incorporated via **taxonomy** or **character data** (total-evidence)



AT**A**T...

TCAC...

????...

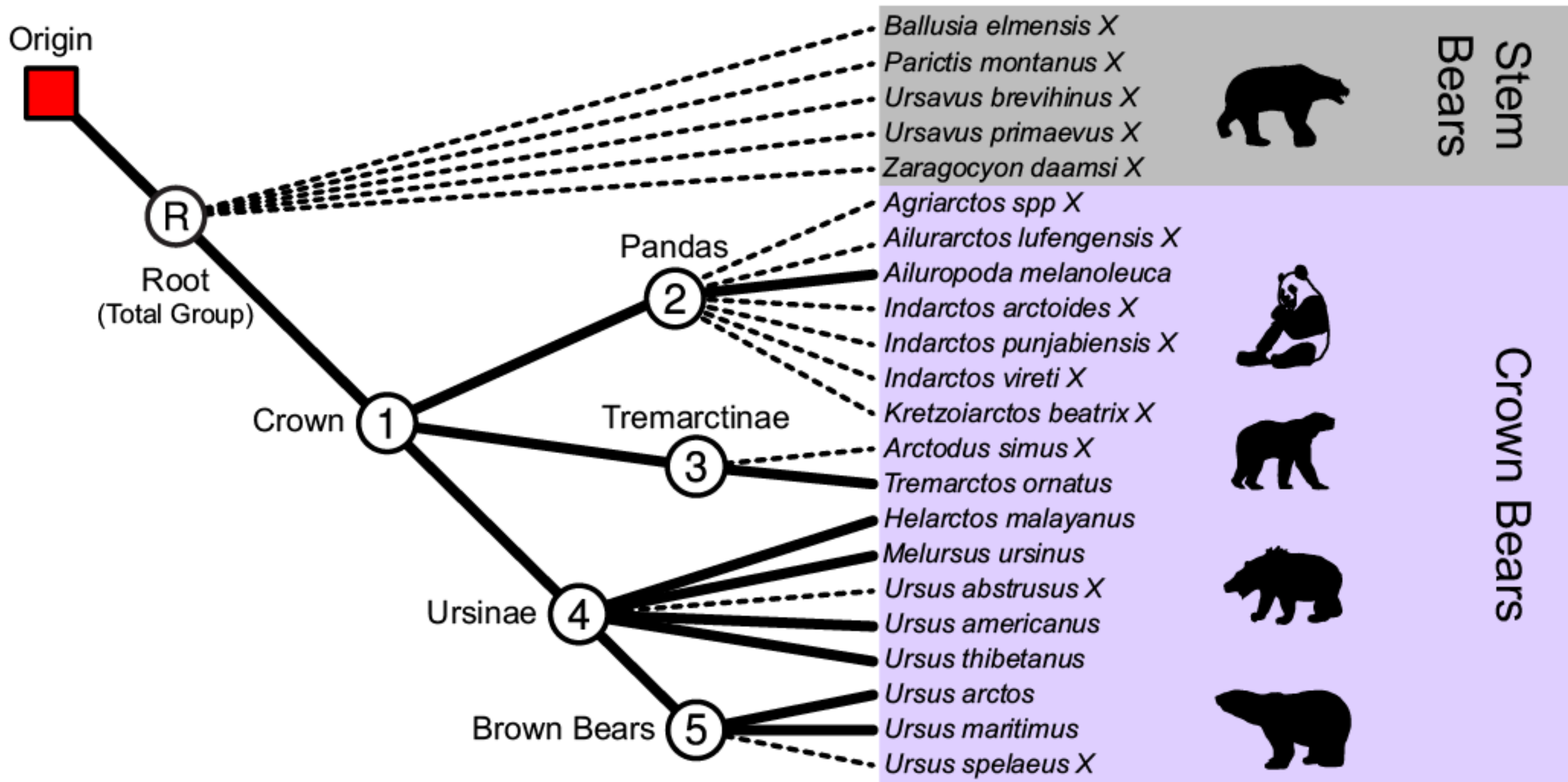
OR

1001...

1101...

0100...

Image source Soul & Friedman ([2015](#))



Data

2 genes (irbp, cytb) for 8 extant species

Occurrence times of 14 fossil species

Taxonomic constraints

Note: this analysis includes includes **no character data** for our fossils

More about models of morphological evolution / TE dating on Thu / Fri

←

→

Search

≡ bears_cytb_fossils.nex

≡ bears_irbp_fossils.nex ×

Users > warnock > Desktop > org > 02_events > TtB > FBD_tutorial > ≡ bears_irbp_fossils.nex

1

#NEXUS

2

3

begin data;

4

dimensions ntax=22 nchar=1281;

5

format datatype=dna missing=? gap=-;

6

matrix

7

Ailuropoda_melanoleuca_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

8

Tremarctos_ornatus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

9

Helarctos_malayanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

10

Melursus_ursinus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

11

Ursus_americanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

12

Ursus_arctos_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

13

Ursus_thibetanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

14

Ursus_maritimus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

15

Parictis_montanus_36.6 ??

16

Zaragocyon_daamsi_21.9 ??

17

Ballusia_elmensis_14.0 ??

18

Ursavus_primaevus_14.4 ??

19

Ursavus_brevihinus_16.2 ??

20

Indarctos_vireti_8.7 ??

21

Indarctos_arctoides_9.6 ??

22

Indarctos_punjabiensis_5.0 ??

23

Ailurarctos_lufengensis_7.7 ??

24

Agriarctos_spp_5.0 ??

25

Kretzoiarctos_beatrix_11.7 ??

26

Ursus_abstrusus_4.3 ??

27

Ursus_spelaeus_0.06 ??

28

Arctodus_simus_0.5 ??

29

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30

end;

31

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Restricted Mode

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UTF-8

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Plain Text

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Add Morphological Data

Add Continuous Geography

Add Discrete Trait

Import Alignment

Template>

Set working dir>

Manage Packages

Clear Class Path

Launch Apps

Save⌘ S

Save As

BEAUti 2: Standard

PartitionsTip DatesSite ModelClock ModelPriorsMCMC

Link Site ModelsUnlink Site ModelsLink Clock ModelsUnlink Clock ModelsLink TreesUnlink Trees

Name	File	Taxa	Sites	Data Type	Site Model	Clock Model	Tree	Ambiguities	
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Split

BEAST 2 Package Manager

List of available packages for BEAST v2.7.*

Name	Installed	Latest	Dependencies	Link	
BEAST.base	2.7.8	2.7.8			BEAST base
BEAST.app	2.7.8	2.7.8	BEAST.base		BEAST base applications
Babel		0.4.2	BEAST.base, BEAST.app, BEAST...		BABEL = BEAST analysis backing effective linguistics
bacter		3.0.1	BEAST.app, feast, BEAST.base		Bacterial ARG inference.
BADTRIP		2.0.0	BEAST.base		Infer transmission time for non-haplotype data and epi data
BASTA		4.0.0	BEAST.base, BEAST.app		Bayesian structured coalescent approximation
bdmm	2.0.0	2.0.0	BEAST.app, MultiTypeTree, BEAS...		Multitype birth-death model (aka birth-death-migration model)
BDMM-Prime	2.4.0	2.4.0	BEAST.base, BEAST.app, SA		Birth-death Migration Model Reimplementation
BDSKY	1.5.1	1.5.1	BEAST.base, BEAST.app		birth death skyline - handles serially sampled tips, piecewise cons
bdtree		0.0.2	BEAST.base, BEAST.app		Birth-death sequential sampling
BEAST_CLASSIC	1.6.4	1.6.4	BEAST.base, BEAST.app		BEAST classes ported from BEAST 1 in wrappers
BEASTLabs	2.0.3	2.0.3	BEAST.base, BEAST.app		BEAST utilities, such as Script, multi monophyletic constraints
BEASTvntr		0.2.0	BEAST.base, BEAST.app		Variable Number of Tandem Repeat data, such as microsatellites
BICEPS		1.1.2	BEAST.app, BEAST.base		Bayesian Integrated Coalescent Epoch PlotS + Yule Skyline
bModelTest		1.3.3	BEAST.base, BEAST.app		Bayesian model test for nucleotide subst models, gamma rate het
BREAK_AWAY		1.2.0	BEASTLabs, BEAST.app, GEO_S...		break-away model of phylogeography
CA		2.1.0	BEAST.base, BEAST.app		Bayesian estimation of clade ages based on probabilities of fossil
CCD	1.0.3	1.0.3	BEAST.app, BEAST.base		Sensible tree set summary/point estimates.
ClaDS		2.1.2	BEAST.base, BEAST.app		Implementation of the ClaDS birth-death tree prior with sampled a
CoalRe		1.0.4	BEAST.base, BEAST.app		Inference of Recombination networks
CodonSubstModels		2.0.0	BEAST.app, BEAST.base		Codon substitution models for DNA
contacTrees		1.2.0	BEAST.app, BICEPS, feast, BEAS...		Phylogenetic model with horizontal transfer for linguistics
contraband		1.0.2	BEAST.base, BEAST.app		Scalable brownian models for continuous trait evolution

☒ Latest

Install/Upgrade

Uninstall

Package repositories

?

Close

Bayesian divergence time estimation

The data

and / or

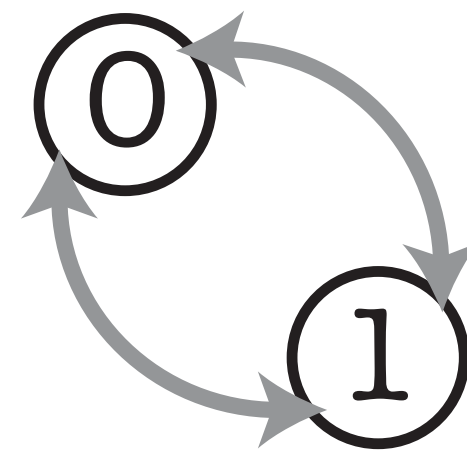
0101... ATTG...
1101... TTGC...
0100... ATTC...



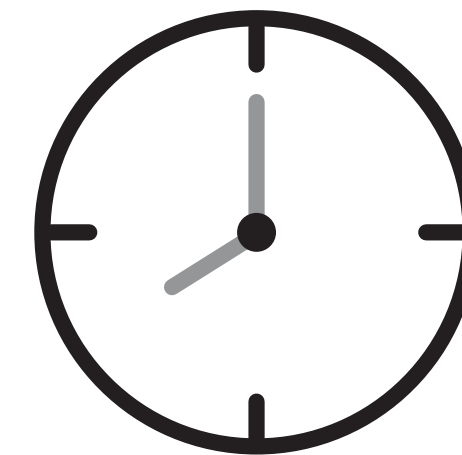
phylogenetics
characters

sample
ages

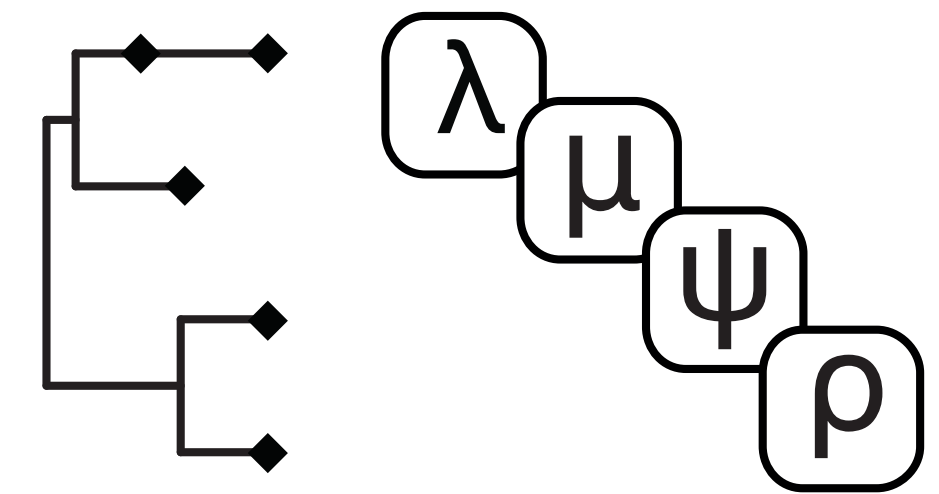
3 model components



substitution
model



clock
model



tree and tree
model

BEAUi 2: Standard

Partitions Tip Dates Site Model ? Clock Model Priors MCMC

Partition

- bears_cytb_fossils
- bears_irbp_fossils
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Gamma Site Model ?

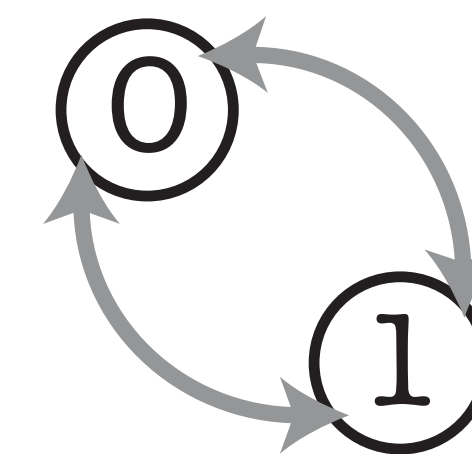
Substitution Rate 1.0 ? e ☐ estimate

Gamma Category Count 0 ?

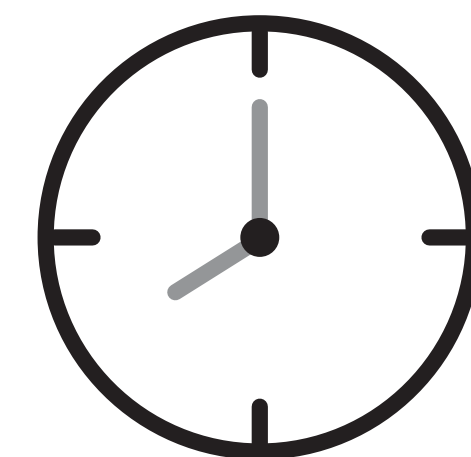
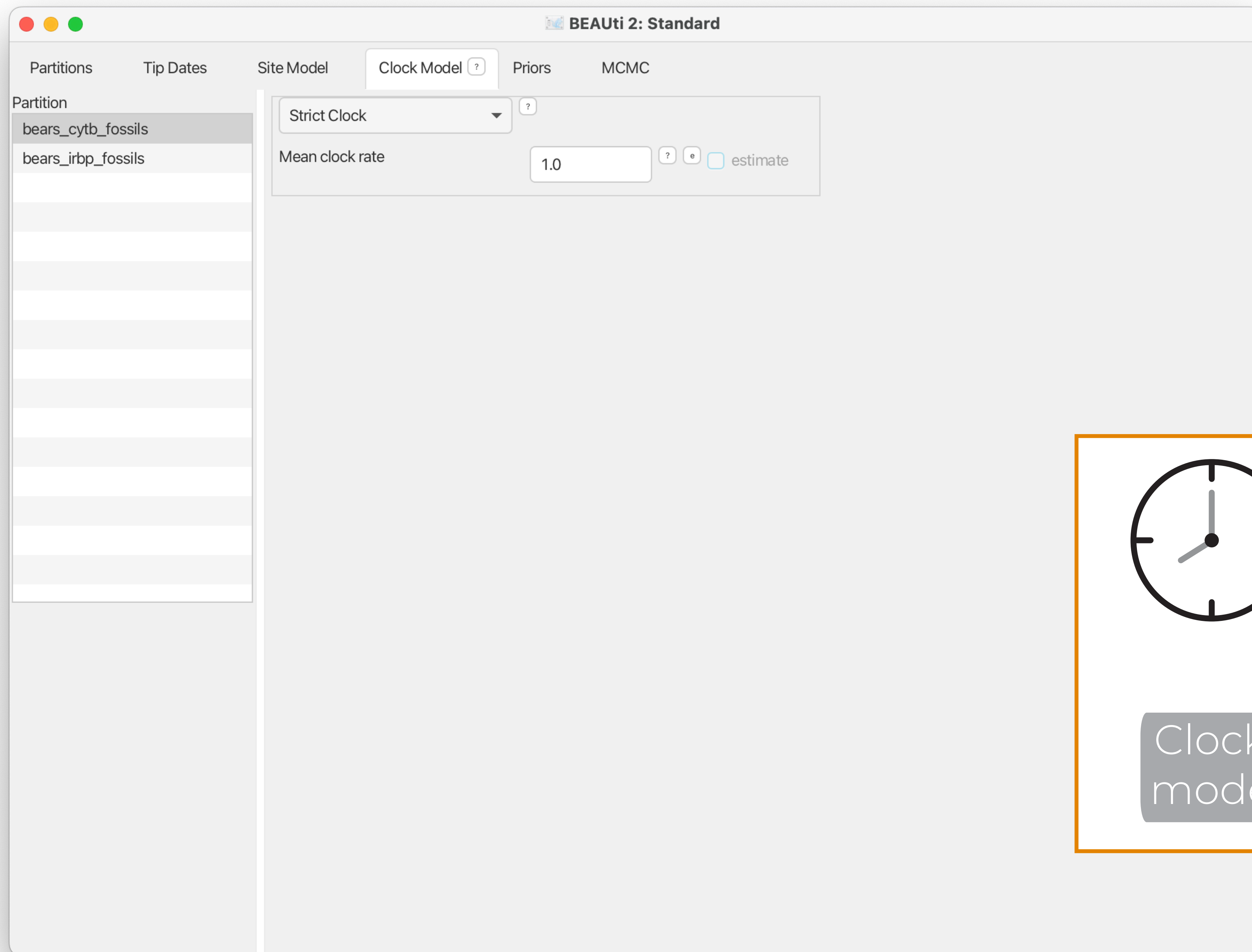
Proportion Invariant 0.0 ? e ☐ estimate

Subst Model JC69 ?

☐ Fix mean substitution rate



Substitution
model



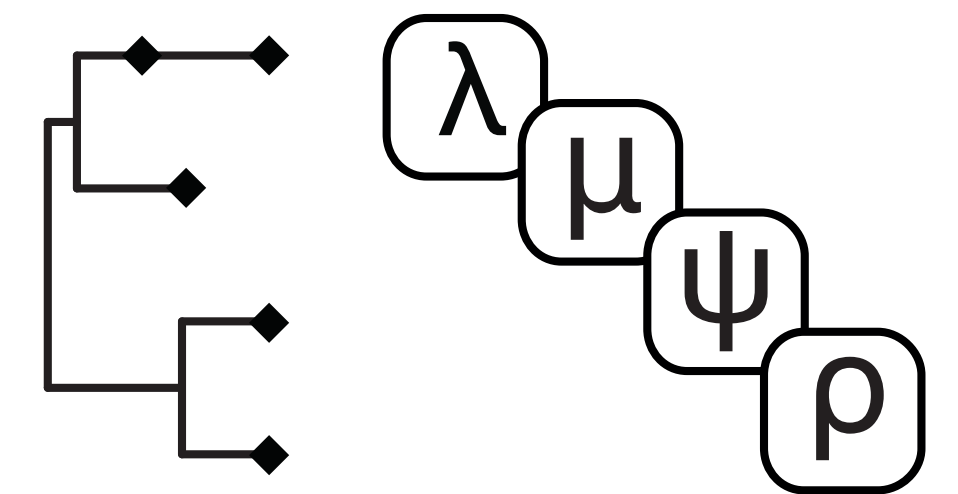
Clock
model

BEAUi 2: Standard

Partitions Tip Dates Site Model Clock Model Priors ? MCMC

▶	Tree.t:bears_cytb_fossils	Fossilized Birth Death Model		
▶	Tree.t:bears_irbp_fossils	Yule Model	?	
▶	birthRate.t:bears_irbp_fossils	Uniform[0.0,Infinity]	?	initial = [1.0] [0.0,∞] Prior on Yule birth rate for partition s:bears_irbp_fossils
▶	diversificationRateFBD.t:bears_cyt...	Uniform[0.0,Infinity]		initial = [1.0] [0.0,∞] prior on diversificationRateFBD.t:bears_cytb_fossils
▶	originFBD.t:bears_cytb_fossils	Uniform[0.0,Infinity]		initial = [100.0] [0.0,∞] prior on originFBD.t:bears_cytb_fossils
▶	samplingProportionFBD.t:bears_cy...	Uniform[0.0,1.0]		initial = [0.5] [0.0,1.0] prior on samplingProportionFBD.t:bears_cytb_fossils
▶	turnoverFBD.t:bears_cytb_fossils	Uniform[0.0,1.0]		initial = [0.5] [0.0,1.0] prior on turnoverFBD.t:bears_cytb_fossils
▶	1CrownBea...	[none]	?	☒ monophyletic -
▶	2Pandas.pr...	[none]	?	☒ monophyletic -
▶	3Tremarcti...	[none]	?	☒ monophyletic -
▶	4Ursinae.p...	[none]	?	☒ monophyletic -

+ Add Prior

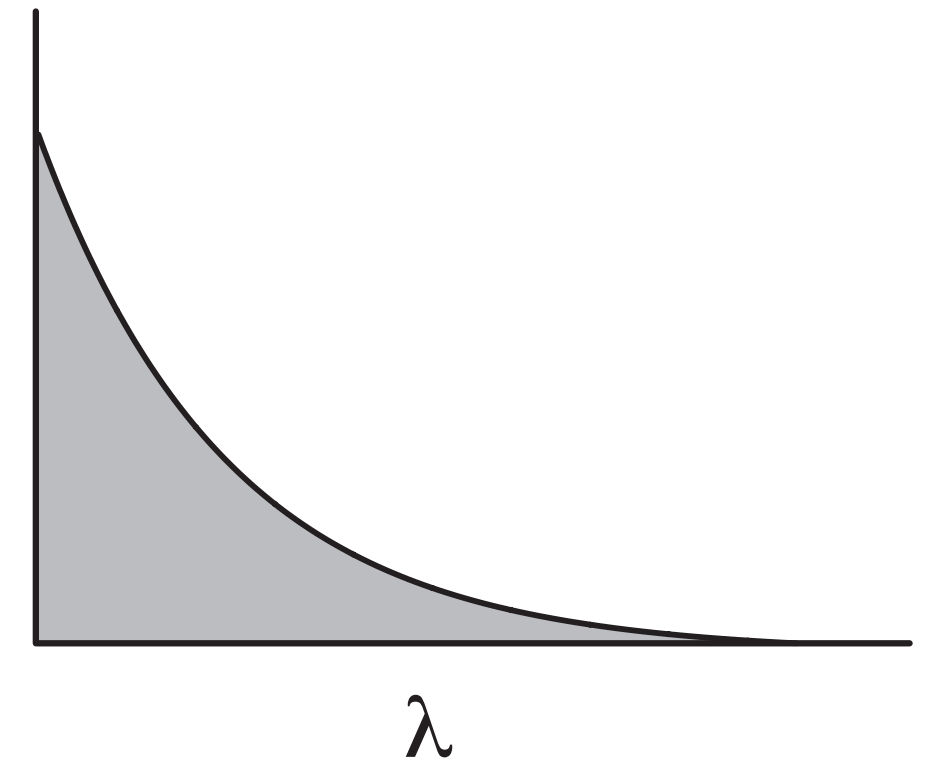


Tree and tree model

Key model parameters

- birth (speciation) rate — λ
- death (speciation) rate — μ
- fossil sampling rate — ψ
- extant sampling probability — ρ

birth, death and sampling are instantaneous rates and typically estimated during inference



extant species sampling is often fixed because this is the parameter we have the most information about

Running analysis “under the prior”

posterior

$$P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \begin{array}{c} \text{0} \rightarrow \text{1} \\ \text{clock} \end{array} \mid \begin{array}{c} \text{0101...} \\ \text{1101...} \\ \text{0100...} \end{array} \mid \text{spiral}\right) =$$

likelihood

probability of the
time tree

priors

$$P\left(\begin{array}{c} \text{0101...} \\ \text{1101...} \\ \text{0100...} \end{array} \mid \begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \begin{array}{c} \text{0} \rightarrow \text{1} \\ \text{clock} \end{array}\right) P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \text{spiral}\right) P(\lambda, \mu, \psi, \rho) P(\text{0} \rightarrow \text{1}) P(\text{clock})$$

$$P\left(\begin{array}{c} \text{0101...} \\ \text{1101...} \\ \text{0100...} \end{array} \mid \text{spiral}\right) \quad \text{marginal pr of the data}$$

Running analysis “under the prior”

$$P\left(\begin{array}{c} \text{graph with } \lambda, \mu, \psi, \rho \\ \text{cycle with 0, 1} \\ \text{clock} \end{array} \middle| \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \text{spiral}\right) =$$

~~$$P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \middle| \begin{array}{c} \text{graph with } \lambda, \mu, \psi, \rho \\ \text{cycle with 0, 1} \\ \text{clock} \end{array}\right) P\left(\begin{array}{c} \text{graph with } \lambda, \mu, \psi, \rho \\ \text{cycle with 0, 1} \\ \text{clock} \end{array} \middle| \text{spiral}\right) P\left(\begin{array}{c} \lambda \\ \mu \\ \psi \\ \rho \end{array}\right) P\left(\begin{array}{c} 0 \\ 1 \end{array}\right) P(\text{clock})$$~~

$$P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \middle| \text{spiral}\right)$$

Parameterisation of the process

We can put priors on different combinations of parameters

Parameter	Transformation
Net diversification	$d = \lambda - \mu$
Turnover	$v = \mu/\lambda$
Sampling proportion	$s = \psi/(\mu + \psi)$
Speciation	$\lambda = d/(1 - v)$
Extinction	$\mu = (vd)(1 - v)$
Sampling	$\psi = (s/(1 - s))((vd)/(1 - v))$

Take care when interpreting PP for
trees with SAs

More options available via the pdf!

e.g., alternative parameterisations, fully extinct clades

Extra slides

Scenario	Software
Node dating with large datasets	MCMCTree
If want (or have to) fix the tree topology	MCMCTree
If fossil sampling is sparse or complex	MCMCTree
If you have abundant fossil data, or are interested in the topological position of fossils	BEAST2, RevBayes
If you're interested in the phylodynamic parameters	BEAST2, RevBayes
If you want to use a specific model	BEAST2, RevBayes, MCMCTree

Table 2. Available fossilized birth–death (FBD) models and extensions

Model name	Description	Reference
FBD specimen process	Model assumes constant rates of diversification (λ , μ) and sampling through time (ψ) and assumes uniform sampling of extant taxa.	Stadler (2010); Heath et al. (2014)
FBD skyline	Model allows for diversification (λ_t , μ_t) and sampling rates (ψ_t) to vary through time across discrete intervals (i.e., in a piecewise-constant fashion).	Stadler et al (2012); Gavryushkina et al. (2014)
Diversified sampling	Model accounts for a non-uniform sampling strategy that aims to maximize the phylogenetic diversity represented in the extant tree. This is achieved by introducing the parameter x_{cut} , which represents the youngest node age, after which we have no more sampled nodes.	Höhna et al. (2011); Zhang et al. (2016)
Multi-type birth–death process	Model allows for variation in diversification and fossil sampling rates across different parts of the tree associated with different “types”. Types may correspond to any trait that effects differences in rates. Lineages can switch between types with rate m .	Kühnert et al. (2016); Barido-Sottani et al. (2020c)
Occurrence birth–death process	Model allows for the incorporation of data from fossil observations outside the tree, i.e., those for which we have no taxonomic information or morphological data, such as trace fossils. These observations are modeled using a separate sampling process, with rate parameter ω .	Manceau et al. (2021); Gupta et al. (2020); Andréoletti et al. (2022)
Episodic FBD process	Model allows for instantaneous speciation, extinction, or sampling events to affect the entire tree, corresponding to events such as mass extinctions or species radiations.	Magee and Höhna (2021)
FBD multispecies coalescent	Model allows separate genes to evolve independently under a coalescent model.	Ogilvie et al. (2022)
FBD range process	Model provides the framework to explicitly take stratigraphic range data into account, thereby associating multiple occurrences within a range of the same taxon.	Stadler et al. (2018)

Further reading

More on FBD models and phylodynamics

From fossils to phylogenies: exploring the integration of paleontological data into Bayesian phylogenetic inference. *Mulvey et al.* ([2025](#))

Integrating fossil observations into phylogenetics using the fossilized birth-death model. *Wright et al.* ([2022](#))

Phylogenetic insights into diversification. *Morlon et al.* ([2024](#))

Unifying birth-death models in epidemiology and macroevolution. *MacPherson et al.* ([2022](#))

decodinggenomes.org (free pdf available) *Stadler et al.* (2024) Chap 9, Phylodynamics

Models of morphological evolution

A systematist's guide to estimating Bayesian phylogenies from morphological data. *Wright* ([2019](#))

Assessing the adequacy of morphological models using posterior predictive simulations. *Mulvey et al.* ([2024](#))